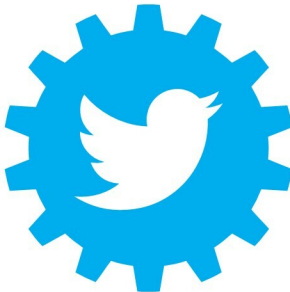


Explore twitter Data Using R

As of August 2017, Twitter had 328 million active users, with 500 million tweets being sent every day. Let's look at how the open source R programming language can be used to analyse the tremendous amount of data created by this very popular social media tool.



Social networking websites are ideal sources of Big Data, which has many applications in the real world.

These sites contain both structured and unstructured data, and are perfect platforms for data mining and subsequent knowledge discovery from the source. Twitter is a popular source of text data for data mining. Huge volumes of Twitter data contain many varieties of topics, which can be analysed to study the trends of different current subjects, like market economics or a wide variety of social issues. Accessing Twitter data is easy as open APIs are available to transfer and arrange data in JSON and ATOM formats.

In this article, we will look at an R programming implementation for Twitter data analysis and visualisation. This will give readers an idea of how to use R to analyse Big Data. As a micro blogging network for the exchange and sharing of short public messages, Twitter provides a rich repository of different hyperlinks, multimedia and hashtags, depicting the contemporary social scenario in a geolocation. From the originating tweets and the responses to them, as well as the retweets by other users, it is possible to implement opinion mining over a subject of interest in a geopolitical location. By analysing the favourite counts and the information about the popularity of users in their followers' count, it is also possible to make a weighted statistical analysis of the data.

Start your exploration

Exploring Twitter data using R requires some preparation. First, you need to have a Twitter account. Using that account, register an application into your Twitter account from <https://apps.twitter.com/> site. The registration process requires basic personal information and produces four keys for R application and Twitter application connectivity. For example, an application *myapptwitterR1* may be created as shown in Figure 1.

In turn, this will create your application settings, as shown in Figure 2.

A customer key, a customer secret, an access token and the access token secret combination forms the final authentication using the *setup_twitter_oauth()* function.

```
>setup_twitter_oauth(consumerKey, consumerSecret,AccessToken,
AccessTokensecret)
```

It is also necessary to create an object to save the authentication for future use. This is done by *OAuthFactory\$new()* as follows:

```
credential<- OAuthFactory$new(consumerKey, consumerSecret,
requestURL, accessURL,authURL)
```